LESSON PLAN

FOLG (26)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Engine Health monitoring by using vibration analysis
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Engine Health monitoring by using vibration analysis
5. Specific Objectives	26.1.1.7 Trainee will be able to explain vibration analysis EHM Technique

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Vibration Spectra Charts; Instructor supervision required for interpretation of frequency bands
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary purpose of Engine Health Monitoring?
(b) List three common methods used for engine condition monitoring.
(c) Define the term 'Vibration Analysis' in the context of EHM.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fundamentals of Vibration Analysis • Sources of Engine Vibration • Frequency Domain Analysis • Time Domain Parameters • Thresholds and Alarm Levels	L & Ds	PPT, IB	15 min
(b)	• Interpretation of Spectra Charts • Identification of Imbalance and Misalignment • Detection of Bearing Defects • Analysis of Gear Mesh Frequencies • Trend Analysis for Predictive Maintenance	L & Ds	PPT, PDF	20 min
(c)	• Integration with EHM Systems • Data Logging and Reporting Procedures • Limitations of Vibration Analysis • Correlation with Power Performance Checks • Role of Maintenance Technicians	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how vibration signatures indicate specific engine faults.
(b) Describe the relationship between vibration data and power performance checks.
(c) Outline the steps involved in reporting vibration anomalies to maintenance.

12. Home Assignment and Follow Up	Prepare a brief note on how vibration analysis data supports the transition to LP 09 regarding Power Performance Checks.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Engine Health monitoring by using Power performance Check	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min